

Customer Agent OS Lite

Fixed Product Features, Subparts, and Sequence Scope

Product definition document - business and user experience scope only

Purpose. Provide a stable, readable product foundation before technical architecture, framework selection, or implementation planning.

Project. Customer Agent OS Lite - a lightweight platform for building and operating domain-specific customer-facing AI agents, proven first through an e-commerce support agent.

Status. Product planning baseline. Changes should be intentional and documented rather than introduced casually during engineering.

Prepared for: Product analysis, portfolio planning, and principal-level engineering design

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1. Product definition and frozen scope

Product definition

Customer Agent OS Lite is a multi-tenant platform for creating, deploying, operating, evaluating, and improving customer-facing AI agents. The first complete application is an e-commerce customer support agent.

The product is intentionally built in a sequence. We first solve real customer journeys, then extract reusable platform capabilities from those journeys. This prevents the project from becoming an abstract agent builder with no proof that its abstractions work in a real business process.

What is fixed

- The product name and positioning: Customer Agent OS Lite.
- The first domain: e-commerce customer support.
- The ten product sequences and the dependency order between them.
- The main user groups, product surfaces, and core product objects.
- The initial customer journeys, human handoff model, quality controls, and governance boundaries.
- The rule that high-risk business actions require explicit confirmation, policy validation, and, where configured, human approval.

What this document does not decide

- Programming languages, frameworks, cloud services, databases, queues, or deployment topology.
- Detailed API contracts or data models.
- Exact implementation of retrieval, orchestration, tracing, or model routing.
- Final visual design system and pixel-level screen layouts.

2. Users, product surfaces, and core objects

2.1 Fixed users

User	Primary purpose
End customer	Receives help, reviews proposed actions, confirms consequential actions, and requests human support.
Human support representative	Takes over difficult or sensitive cases, approves actions, resolves exceptions, and supplies corrective feedback.
Support operations manager	Configures journeys, policies, knowledge, quality rules, and operational performance.
Developer or integration engineer	Builds tools, domain extensions, integrations, tests, and reusable agent capabilities.
Tenant administrator	Controls users, roles, security, environments, model access, budgets, and retention settings.

2.2 Fixed product surfaces

Surface	What it contains
Customer support interface	Web chat, conversation history, streaming replies, file upload, confirmations, progress, sources, and handoff.
Human support workspace	Escalation inbox, case context, transcript, AI summary, approval actions, human takeover, and resolution feedback.
Agent Studio	Agent, journey, knowledge, tool, policy, model, evaluation, release, and deployment management.
Developer platform	SDKs, MCP templates, configuration as code, local runner, APIs, webhooks, mocks, and documentation.
Quality and operations center	Evaluation scorecards, trace explorer, replay, failure analysis, cost, latency, support metrics, and alerts.
Administration and governance center	Organizations, environments, roles, permissions, approvals, audit history, retention, allowlists, and quotas.

2.3 Core product objects

Object	Meaning
Organization	A company using the product.
Environment	Development, staging, or production.
Agent	A customer-facing AI agent with a defined purpose and boundaries.
Domain pack	Reusable journeys, terminology, tools, policies, knowledge patterns, tests, and dashboards for an industry.
Journey	A customer goal the agent can complete or hand off.
Knowledge source	Approved information used to ground answers and decisions.
Tool	A permitted business action or data lookup.
Policy	A rule that constrains behavior, eligibility, approvals, or access.
Conversation	A customer-agent interaction across one or more turns.
Workflow	A multi-step process that may pause and resume.
Human queue	A destination for escalated or approval-required work.
Evaluation suite	Tests that measure quality, safety, cost, latency, and outcomes.
Release	An immutable version of an agent and its related configuration.
Audit event	A reconstructable record of an important decision or action.

3. Sequence 1 - E-commerce reference agent

Product boundary	Fixed decision
Goal	Build one useful customer-support agent before attempting to build a platform for many agents.
Definition of done	Every supported journey either completes successfully or escalates with enough context that the customer does not need to repeat the issue.
Not included in this sequence	Voice, email, WhatsApp, multiple industries, visual agent building, and fully autonomous high-value refunds.

3.1 Fixed customer journeys

Journey	Fixed capability
Order tracking	Identify the customer and order; retrieve shipment state; explain delay or expected delivery; escalate abnormal situations.
Return eligibility	Find the order; identify the item and applicable policy; calculate eligibility; explain the decision with supporting source.
Create a return	Select item and reason; select refund or credit; preview the action; confirm; create authorization; provide label and instructions.
Damaged-item replacement	Collect description and evidence; verify order; check policy and inventory; request approval when needed; create replacement or escalate.
Refund request	Determine eligibility and amount; show a clear preview; collect confirmation; perform the action; report only a verified result.
Subscription cancellation	Show current plan and consequences; offer at most one permitted retention option; confirm timing; cancel and provide proof.
Human escalation	Transfer when requested, low-confidence, high-risk, failed, unsupported, or repeatedly disputed; preserve complete context.

3.2 Shared customer experience features

- Web chat with streaming responses and conversation continuity.
- Customer identification and authentication when account-specific data or actions are needed.
- Clarifying questions when the request is ambiguous or required information is missing.
- Policy citations and visible evidence for eligibility or policy explanations.
- Action previews showing what will change before the customer confirms.
- Explicit confirmation for consequential actions.
- Workflow progress and clear next steps for work that cannot finish immediately.
- Graceful failure messages that state what failed and what will happen next.
- Human handoff without forcing the customer to restart the explanation.

4. Sequence 2 - Human support workspace

Product boundary	Fixed decision
Goal	Make human assistance a first-class part of the product instead of treating escalation as an error state.
Definition of done	A representative can understand the case within seconds, take over, approve or reject actions, resolve the issue, and feed corrections back into quality improvement.
Not included in this sequence	A complete CRM, workforce scheduling, payroll, call-center staffing, and broad enterprise case management.

4.1 Escalation inbox

- New and assigned escalations.
- Urgency, sentiment, journey type, waiting time, risk level, and queue.
- Filters for approval requests, system failures, customer-requested handoff, and policy exceptions.

4.2 Complete handoff context

- Customer identity and relevant account context.
- Conversation transcript and concise AI-generated summary.
- Customer request, current journey, and current workflow state.
- Knowledge and policies consulted.
- Tools called, results, failures, and pending actions.
- Reason for escalation and recommended next action.

4.3 Human actions

- Join the conversation and take full control.
- Approve or reject a pending action.
- Edit a draft response or complete a manual business action.
- Return control to the agent after resolving an exception.
- Close the case with a reason and outcome.
- Classify an agent mistake as wrong answer, wrong tool, missing knowledge, policy issue, or incorrect escalation.

5. Sequence 3 - Reusable agent runtime

Product boundary	Fixed decision
Goal	Turn the e-commerce agent from a hard-coded application into an agent running on reusable product primitives.
Definition of done	The e-commerce agent is described through agent, journey, tool, knowledge, policy, and runtime definitions; a second simple agent can be created without changing the core runtime.

5.1 Agent definition

- Name and purpose

- Domain and brand identity
- Tone and supported languages
- Supported journeys
- Connected tools and knowledge
- Applied policies
- Approved models
- Human queues
- Usage and execution limits

5.2 Journey definition

- Customer goal
- Entry conditions
- Required information
- Permitted tools
- Relevant policies
- Completion conditions
- Escalation conditions
- Success metrics

5.3 Context and memory

- Assemble only the current conversation, customer context, journey state, tool results, relevant knowledge, and tenant rules needed for the current task.
- Separate current-session memory, durable customer facts, previous support history, active workflow state, and temporary tool results.
- Do not store every conversational detail as permanent memory.

5.4 Registries and runtime controls

- Tool registry with purpose, permissions, schemas, risk level, environment, and health.
- Policy registry attachable to organization, agent, journey, tool, customer segment, or environment.
- Execution controls for steps, tools, time, cost, retries, cancellation, and escalation.
- Versioned agent changes rather than silent production mutation.

6. Sequence 4 - Knowledge and integration platform

Product boundary	Fixed decision
Goal	Give agents reliable information and safe, governed access to business systems.
Definition of done	Administrators can connect approved knowledge and commerce systems; agents can produce grounded answers and perform safe, verifiable actions.
Not included in this sequence	Unrestricted database queries or arbitrary internal API execution by the model.

6.1 Knowledge platform

Feature group	Fixed subparts
Sources	PDF, Word, Markdown, web pages, FAQ collections, help centers, product documentation, and structured policy data.
Synchronization	Add, sync, show last successful update, detect changes, disable, remove, and retry failed processing.
Versioning	Document version, effective date, expiration date, previous version, owner, last review, and status.
Permissions	Restrictions by organization, agent, product, region, customer segment, employee role, and language.
Grounding	Show source, policy title, relevant passage, effective date, and confidence indicator.
Quality	Detect missing answers, conflicts, stale documents, weak searches, unused knowledge, and escalation-causing gaps.

6.2 Integration platform

- Initial services: customer profile, orders, inventory, shipping, returns, refunds/payments, subscriptions, and support tickets.
- MCP client and registry plus commerce, customer, logistics, and support MCP server templates.
- Tool management showing schemas, permissions, environment, confirmation requirements, risk, health, volume, latency, and errors.
- Action confirmation with preview, customer confirmation, policy check, permission check, verified result, and final receipt.
- Safe retries that prevent duplicate refunds, returns, shipments, subscriptions, or tickets.
- Sandbox mode using simulated systems and data.

7. Sequence 5 - Advanced orchestration and workflows

Product boundary	Fixed decision
Goal	Support simple conversational requests and complex multi-step processes without creating an uncontrolled agent swarm.
Definition of done	A damaged-item replacement can pause for customer evidence, approval, inventory, warehouse, or carrier events and later resume without losing state or repeating completed actions.
Not included in this sequence	Unbounded swarms, unlimited reasoning loops, autonomous creation of new agents, and unapproved cross-organization collaboration.

7.1 Fixed orchestration modes

Mode	Used for	Product behavior
ReAct-style execution	Order lookup, policy clarification, simple	Observe the situation, choose a useful next

Mode	Used for	Product behavior
	troubleshooting, and short dynamic requests.	action, inspect the result, and answer or continue within strict limits.
Plan-and-execute	Replacement, multi-item return, approval-required refund, address changes, and cancellation with retention decisions.	Create a structured plan, validate it, execute and track steps, adjust only when necessary, then complete or escalate.

7.2 Fixed bounded specialists

Specialist	Responsibility
Triage	Intent, journey, urgency, risk, and required capabilities.
Knowledge	Query reformulation, retrieval, conflicts, source selection, and citation preparation.
Customer operations	Orders, returns, refunds, replacements, and subscription changes.
Risk and policy	Eligibility, thresholds, fraud signals, exceptions, and approval requirements.
Handoff	Summary, reason for escalation, completed work, pending work, and recommended next action.

7.3 Durable workflow capabilities

- Pause and wait for customer input
- Wait for human approval
- Wait for carrier, warehouse, or inventory events
- Resume after interruption
- Retry temporary failures
- Cancel safely
- Compensate or reconcile partially completed work

8. Sequence 6 - AI gateway and inference economics

Product boundary	Fixed decision
Goal	Operate multiple models through one controlled product layer while balancing quality, reliability, latency, and cost.
Definition of done	The same agent can use multiple approved models, enforce budgets, survive a provider failure, and report the cost of each completed journey.
Not included in this sequence	Custom foundation-model training, proprietary inference engines, and large self-managed GPU clusters.

Feature group	Fixed subparts
Model catalog	Provider, model, version, capabilities, tool support, structured output,

Feature group	Fixed subparts
	context capacity, expected quality, latency, cost, and approved use cases.
Task routing	Separate routing for classification, retrieval answers, planning, image understanding, summarization, evaluations, and guardrail checks.
Tenant policies	Allowed providers, models, default, restricted tasks, data requirements, and fallback order.
Reliability	Safe retry, approved fallback, deterministic fallback path, human handoff, and no false success claims.
Budgets	Limits by organization, agent, journey, customer, day, month, and individual execution.
Cost reporting	Cost per conversation, journey, model, tenant, success, failure, and escalation.
Efficiency	Conversation compaction, context limits, cache controls, model-call limits, tool-call limits, and offline processing.

Primary economics metric

Cost per successful autonomous resolution.

9. Sequence 7 - Safety, governance, and permissions

Product boundary	Fixed decision
Goal	Ensure agents act only within explicit business, customer, user, and organizational boundaries.
Definition of done	No high-risk action bypasses permission, policy, confirmation, and approval requirements; every consequential action can be reconstructed from audit history.
Not included in this sequence	Claims of formal certifications that the project has not actually obtained.

9.1 Roles and permissions

- Organization administrator
- Security administrator
- Support operations manager
- Support representative
- Developer
- Evaluator
- Read-only auditor

9.2 Guardrail layers

Layer	Fixed checks
Input	Prompt injection, jailbreak, abuse, sensitive information, unsupported requests, malicious files, and input size.
Knowledge	Tenant isolation, document ACL, source validity, effective dates, approved source types, and malicious content.
Planning	Allowed goal, authentication sufficiency, prohibited steps, approvals, budget, and risk.
Tool	Permission, schema, business rules, customer confirmation, idempotency, limits, approval state, and rate limit.
Output	Sensitive-data leakage, unsupported claims, missing evidence, policy conflicts, prohibited promises, brand tone, and clarity.

9.3 Approval and privacy controls

- Approval categories for high-value refunds or replacements, address changes after shipment, account-sensitive changes, policy exceptions, fraud risk, and legal or security complaints.
- Audit history containing initiator, agent version, model, policy result, tool, approval, prior state, new state, result, timestamp, and conversation.
- Configurable retention, deletion, PII redaction, export, logging restrictions, and environment separation.

10. Sequence 8 - Evaluation and release management

Product boundary	Fixed decision
Goal	Make agent quality measurable and prevent unsafe or lower-quality changes from reaching customers.
Definition of done	A new release cannot reach production without passing critical safety, policy, isolation, and journey evaluations; a failed release can roll back to a known version.

10.1 Evaluation assets

- Datasets with customer input, profile, history, knowledge, tool responses, expected action, expected answer properties, and escalation expectations.
- Evaluation categories for intent, retrieval, groundedness, policy, tool choice, tool arguments, workflow completion, escalation, tone, safety, latency, and cost.
- Simulated customer types: normal, ambiguous, angry, impatient, missing-information, changing-requirements, fraudulent, prompt-injection, multilingual, and unsupported request.
- Test levels: component, knowledge, tool, policy, journey, end-to-end simulation, adversarial, and sampled production evaluation.

10.2 Releases

- Every release freezes the agent, journeys, prompts, tool schemas, policies, knowledge references, routing settings, evaluation results, and source revision.
- Fixed stages: Draft -> Development -> Evaluation -> Staging -> Shadow -> Canary -> Production.

- Release comparison, approvals, automated gates, canary percentage, rollback, history, and change notes.
- A failed production conversation can become a labeled regression test for future releases.

11. Sequence 9 - Observability and improvement center

Product boundary	Fixed decision
Goal	Let support, product, and engineering teams understand what the agent did, why it did it, and whether it helped the customer.
Definition of done	An authorized investigator can explain a failed conversation, involved models and tools, policy decisions, retrieved evidence, cost, and whether the issue is isolated or systemic.

Feature group	Fixed subparts
Trace explorer	Customer message, selected agent and journey, context, retrieval, model calls, specialists, policies, tools, approvals, answer, latency, and cost.
Conversation replay	Original input, decisions, evidence, tool outputs, policy results, final answer, and access-based redaction.
Operational metrics	Volume, active conversations, errors, response latency, model and tool latency, queues, provider failures, fallbacks, workflow backlog, and cache effectiveness.
Support metrics	Autonomous resolution, first-contact resolution, escalation, reopen, override, satisfaction, resolution time, and incorrect actions.
Quality metrics	Unsupported answers, policy violations, retrieval failures, tool accuracy, approval correctness, and release comparisons.
Cost metrics	Tokens, calls, conversation cost, journey cost, customer cost, successful-resolution cost, and failed-run waste.
Improvement	Failure clustering, knowledge-gap detection, suggested fixes, and alerts for outages, regressions, cost spikes, failures, backlogs, or budget exhaustion.

12. Sequence 10 - Agent Studio and developer platform

Product boundary	Fixed decision
Goal	Expose only the capabilities proven by the reference agent so other agents and domains can be created safely.
Definition of done	A developer or operations user can create a second agent without modifying the internal runtime.
Not included in this sequence	A free-form visual workflow canvas before the product abstractions have been validated.

12.1 Agent Studio

Area	Fixed subparts
Agent creation	Template, name, purpose, brand, tone, languages, journeys, knowledge, tools, policies, models, budgets, and human queues.
Journey management	Required information, tools, policies, escalation, completion, and example conversation preview.
Knowledge management	Sources, sync, errors, conflicts, authoritative designation, retrieval inspection, and gaps.
Tool management	Registration, schema, permission, confirmation, risk, environment, sandbox test, health, and failures.
Policy management	Return windows, refund limits, approvals, prohibited actions, authentication, and escalation triggers.
Model settings	Approved models, defaults, fallbacks, task routing, token limits, and budgets.
Evaluation and release	Tests, conversation imports, suites, comparisons, failures, regressions, release creation, staging, canary, production, and rollback.

12.2 Developer platform

- Python SDK for agents, journeys, tools, policies, evaluators, specialists, and domain packs.
- TypeScript SDK for embedded chat, authentication, events, files, workflow progress, and human handoff.
- MCP starter kit with authentication, schemas, permission hooks, audit hooks, tests, and local mocks.
- Configuration as code for agent, journey, policy, routing, tool, evaluation, and deployment settings.
- Local development environment with simulated customers, mock commerce systems, traces, replay, evaluations, and failure injection.
- Developer documentation covering quick start, concepts, tools, MCP, evaluations, guardrails, deployments, and troubleshooting.

13. Domain pack, expansion plan, exclusions, and final statement

13.1 Fixed first domain pack

E-commerce Support Pack

Includes the seven journeys, commerce terminology, example policies and knowledge, commerce/customer/logistics/support tools, mock business systems, evaluation datasets, edge cases, dashboards, and escalation rules.

13.2 Post-v1 expansion

Expansion	Product additions
Multimodal claims	Image upload, damage assessment, receipt/document understanding, secure attachments, and image-specific evaluations.
Voice agent	Incoming voice, transcription, speech, interruption handling, transfer, and voice-specific quality checks using the same runtime.

Expansion	Product additions
Additional channels	Email, SMS, WhatsApp, and in-app messaging using the same agent, tools, policies, and workflows.
Proactive support	Delayed-order, payment-failure, renewal, return-status, and outage notifications.
SaaS domain pack	Account access, billing, plan changes, API troubleshooting, integration failure, and product-bug escalation.

13.3 Explicit v1 exclusions

- Proprietary foundation model or custom vector database.
- Large GPU infrastructure or multi-cloud GPU scheduling.
- Autonomous swarms of many agents.
- Every customer-service industry and channel.
- Complete contact-center, CRM, workforce, or generic automation replacement.
- Fully autonomous high-risk financial actions.
- Automatic self-modification in production.
- Formal compliance certification claims.

13.4 Frozen sequence

1. Build the e-commerce support agent.
2. Build the human support workspace.
3. Extract the reusable agent runtime.
4. Build knowledge and integration capabilities.
5. Add advanced orchestration and durable workflows.
6. Add the AI gateway and inference economics.
7. Add safety, governance, and permissions.
8. Add evaluations and release management.
9. Add observability and improvement tooling.
10. Expose proven capabilities through Agent Studio and SDKs.

Final product statement

Customer Agent OS Lite is a lightweight, production-oriented platform for building and operating domain-specific customer-facing AI agents. It unifies customer conversations, approved knowledge, business actions, human support, evaluation, governance, release management, and operational intelligence. Its first complete implementation is an e-commerce support agent for orders, returns, refunds, damaged-item replacements, subscription management, and human escalation.

Scope control

This document is the fixed feature baseline. The next product step is the V1 Product Contract, which turns the first journeys, personas, outcomes, approval boundaries, screens, and acceptance criteria into precise decisions for technical design.